SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – CLASS TEST (5 Questions)

Course Code:	DSA06	**Course Title:**	Data Handling and Visualization
CO Assessed:	CO5 – Precision (BL6)	**Assessment:**	Assessment Tool 1 – Class Test
Weightage:	40% of CIA	**Total Marks:**	50 (5 Questions × 10 Marks)
CO5 Statement:	*Create effective and interactive visualizations by integrating data wrangling and advanced visualization tools (BL6)*		
Student Name:	RENEE VINCENT ASR	**Reg. No.:**	192424161
Section / Batch:	BTECH AI & DS	**Date:**	14-08-2026

Code of Conduct

I Renee Vincent ASR-192424161 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Renee Vincent

Instructions

- This test contains 5 questions, each carrying 10 marks. Total: 50 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Uncertainty & Confidence Intervals	Uncertainty of point estimates, confidence intervals, trend estimation	1	10
Handling Overplotting	Overlapping points, jittering, transparency	2	10
Principles of Effective Visualization	Proportional ink principle, avoiding misleading axes	3	10
Accessibility & Misuse of Color	Color accessibility, avoiding misuse of color	4	10
Probability & Hypothetical Outcome Plots	Probability distribution and hypothetical outcome plots	5	10
		GRAND TOTAL:	**50 Marks**

Annexure A – Class Test

1. A survey estimates the average monthly income (₹) of two cities:

City	Mean Income	Confidence Interval
A	25,000	± 2,000
B	27,000	± 5,000

Represent this dataset using a suitable graph to visualize uncertainty. Justify your choice and interpret which city has more reliable estimates.

Aim:

To visualize the mean monthly income of two cities along with their confidence intervals using an **error bar chart** in R.

Suitable Graph:

Error Bar Chart

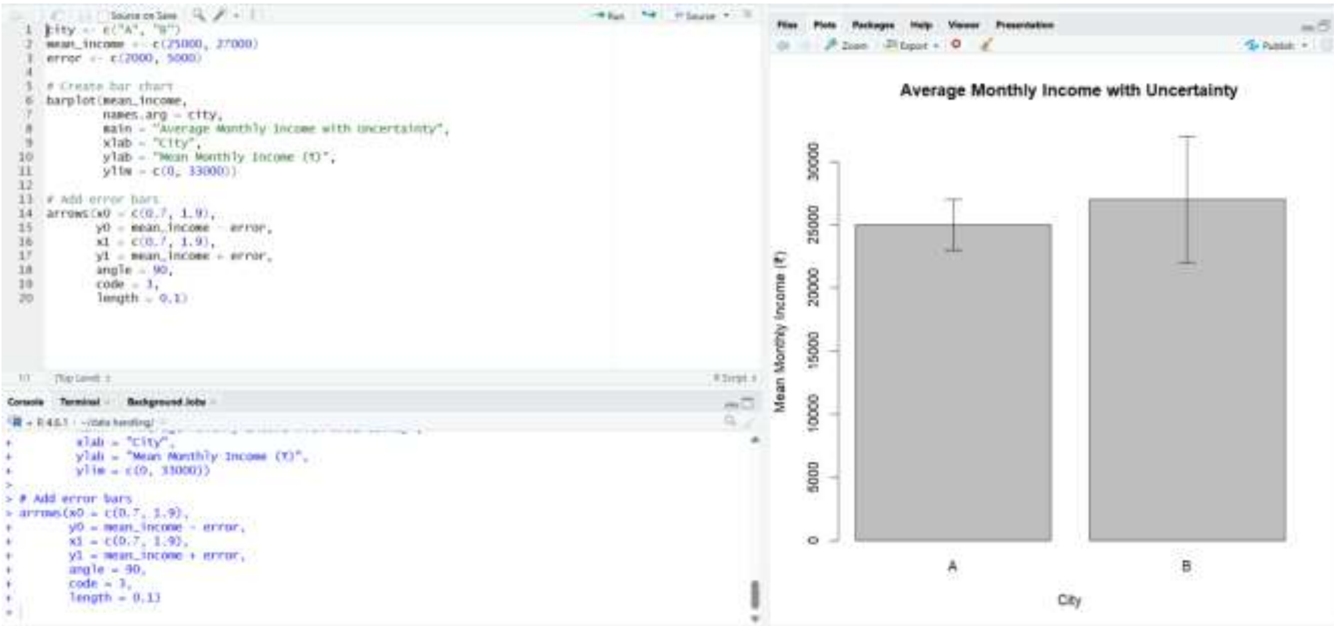
An error bar chart is suitable because it shows:

- The estimated mean income.
- The uncertainty around each estimate.
- The width of the confidence interval.

Procedure:

1. Open **RStudio**.
2. Create a new R script.
3. Enter the city names.
4. Enter the mean incomes.
5. Enter the confidence interval values.
6. Create a bar chart using barplot().
7. Calculate the lower and upper limits using:
    - Lower limit = Mean − Error
    - Upper limit = Mean + Error
8. Use arrows() to display the error bars.
9. Run the program.
10. Observe the graph in the **Plots** window.

Output:



Interpretation:

City B has the higher estimated average income (**₹27,000**), but its estimate has greater uncertainty. City A has the **more reliable estimate** because its confidence interval is narrower (±₹2,000 compared with ±₹5,000).

Justification:

An error bar chart is appropriate because it communicates both the central estimate and its uncertainty. A narrower error bar indicates a more precise estimate.

2. A dataset contains exam scores of 200 students, where many students have similar scores:

Score	Frequency
50	30
60	45
70	60

80	**40**
90	**25**

Choose an appropriate visualization to avoid overplotting. Explain how techniques like jittering or transparency can improve clarity.

Aim:

To visualize the distribution of student exam scores while avoiding overplotting using jittering and transparency.

Suitable Graph:

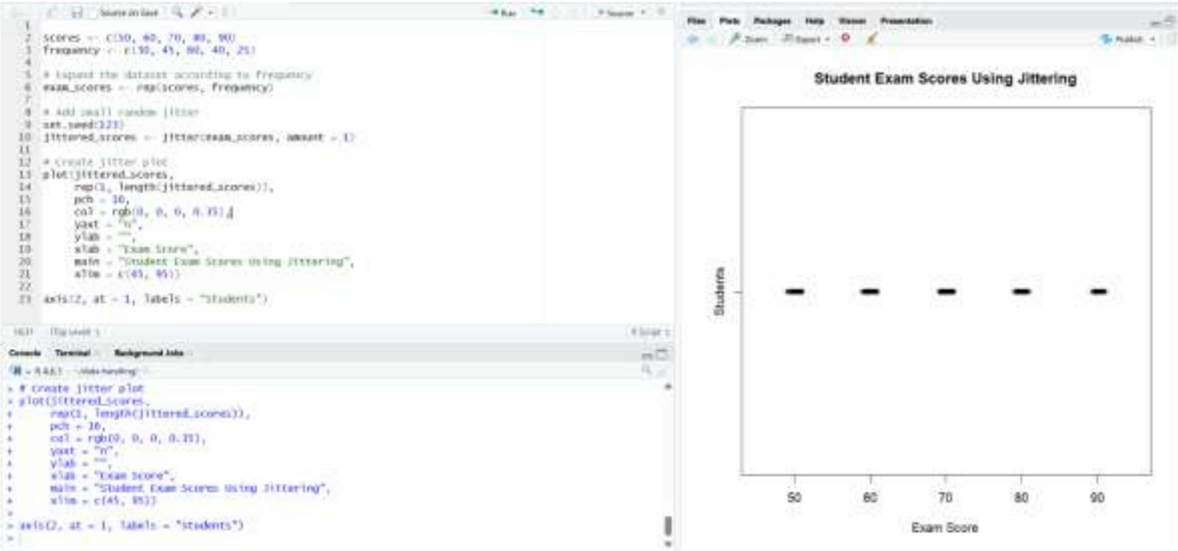
Jitter Plot / Strip Plot

Since many students have identical scores, plotting all points at exactly the same position would cause overplotting. Jittering slightly separates overlapping points.

Procedure:

1. Open RStudio.
2. Enter the score values.
3. Enter their frequencies.
4. Use rep() to create individual observations.
5. Apply jitter() to slightly move overlapping points.
6. Use transparent points to improve visibility.
7. Plot the points.
8. Run the program.
9. Observe the distribution in the Plots window

Output:



Interpretation:

- 50 → 30 students
- 60 → 45 students
- 70 → 60 students
- 80 → 40 students
- 90 → 25 students

The score of **70 has the highest concentration**.

Justification:

A jitter plot is suitable because it reduces overplotting and makes repeated observations visible. Transparency further improves the visualization by showing areas where many points overlap.

3. A company reports revenue growth:

Year	Revenue (₹ lakhs)
2021	100
2022	110
2023	115

Create a graph that avoids misleading axes and follows the proportional ink principle. Explain how improper axis scaling could distort interpretation.

Aim:

To visualize revenue growth using a correctly scaled line chart while following the proportional ink principle and avoiding misleading axis truncation.

Suitable Graph:

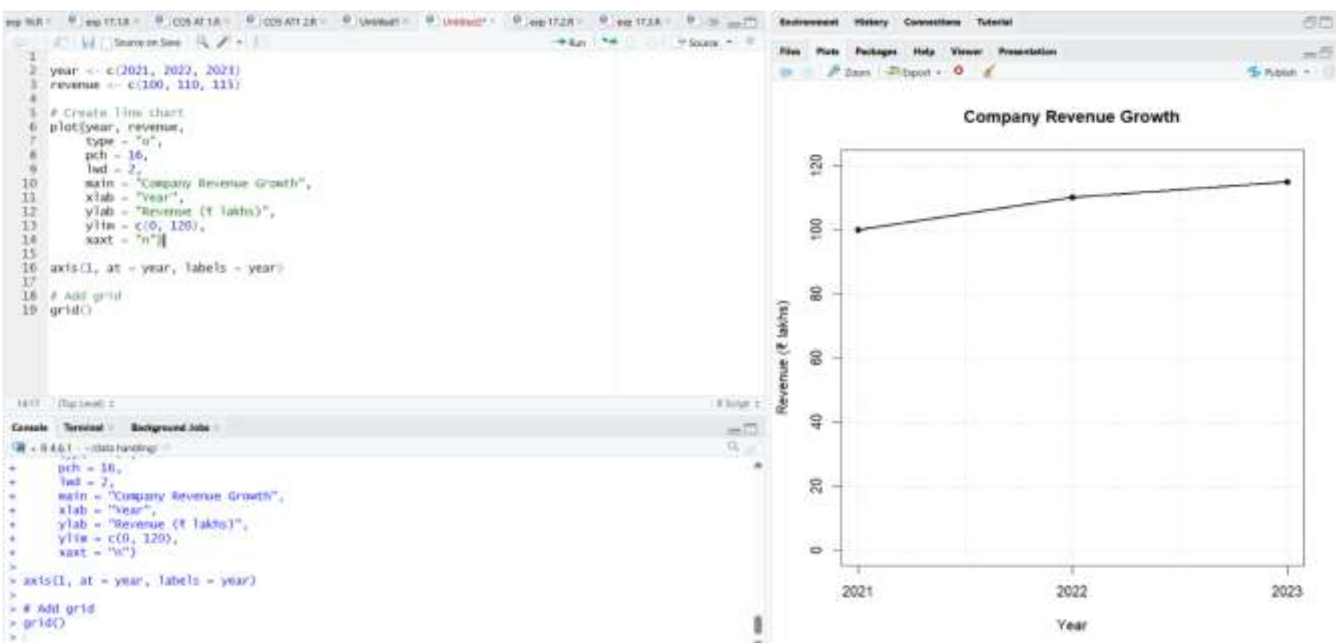
Line Chart with a properly scaled Y-axis

A line chart is suitable because the data represents revenue over time.

Procedure:

1. Open RStudio.
2. Enter the year values.
3. Enter the corresponding revenue.
4. Use plot() to create a line graph.
5. Set the Y-axis to start from **0**.
6. Set a suitable upper limit.
7. Add points and connecting lines.
8. Add labels and title.
9. Run the program.

Output:



Interpretation:

Revenue increased by:

₹115 − ₹100 = ₹15 lakhs

Percentage increase:

15 / 100 × 100 = 15%

Therefore, the company experienced a **15% increase in revenue** over the period.

Justification:

A line chart is appropriate for time-series data. A properly scaled Y-axis prevents visual exaggeration and follows the proportional ink principle.

4. A dataset shows profit/loss across departments:

Department	Status
HR	Profit
Sales	Loss
IT	Profit
Finance	Loss

Design a suitable visualization ensuring accessibility. Suggest color choices and explain how to avoid misuse of color for colorblind users.

Aim:

To visualize departmental profit and loss using an **accessible bar chart** that does not rely only on color.

Suitable Graph:

Bar Chart

A bar chart is suitable for comparing categorical data.

For accessibility, we should not communicate information through color alone.

We can use:

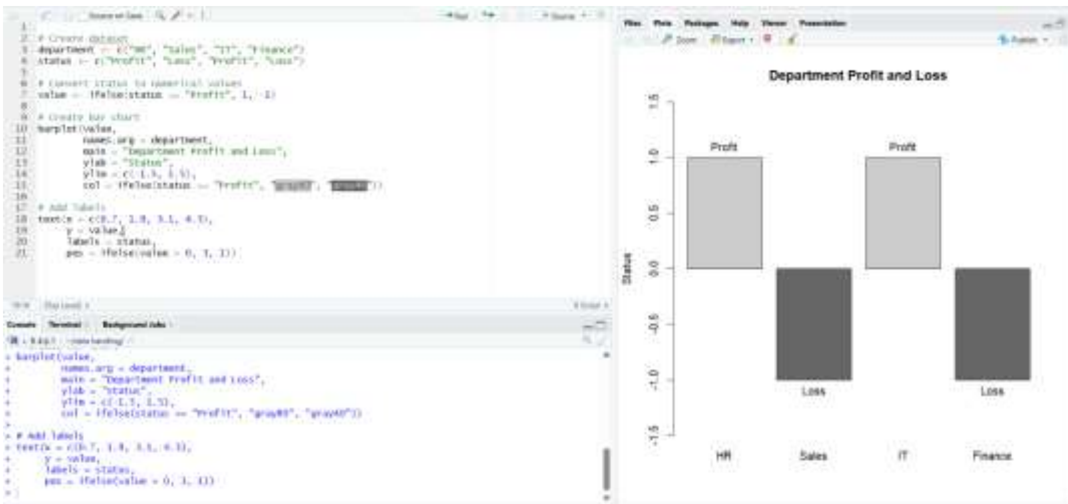
- Different colors
- Different patterns/symbols
- Text labels

Procedure:

1. Open RStudio.
2. Enter department names.
3. Enter the corresponding status.

4. Convert Profit to a positive value.
5. Convert Loss to a negative value.
6. Create a bar chart.
7. Add text labels.
8. Use grayscale shades rather than depending entirely on red and green.
9. Run the program.

Output:



Interpretation:

HR and IT are profitable departments, whereas Sales and Finance have losses.

Justification:

A bar chart clearly compares categorical values. Using position and text labels in addition to color makes the visualization accessible to users with color-vision deficiencies.

5. A factory produces light bulbs with a 5% defect rate. A quality control engineer wants to visualize the probability of defects in batches of 20 bulbs.
Represent the probability distribution using a suitable graph (e.g., hypothetical outcome plot or probability plot). Interpret the likelihood of observing 0, 1, or more defective bulbs.

Aim:

To visualize the probability distribution of defective bulbs in a batch of 20 using a binomial probability distribution.

Suitable Graph:

Binomial Probability Distribution Bar Chart

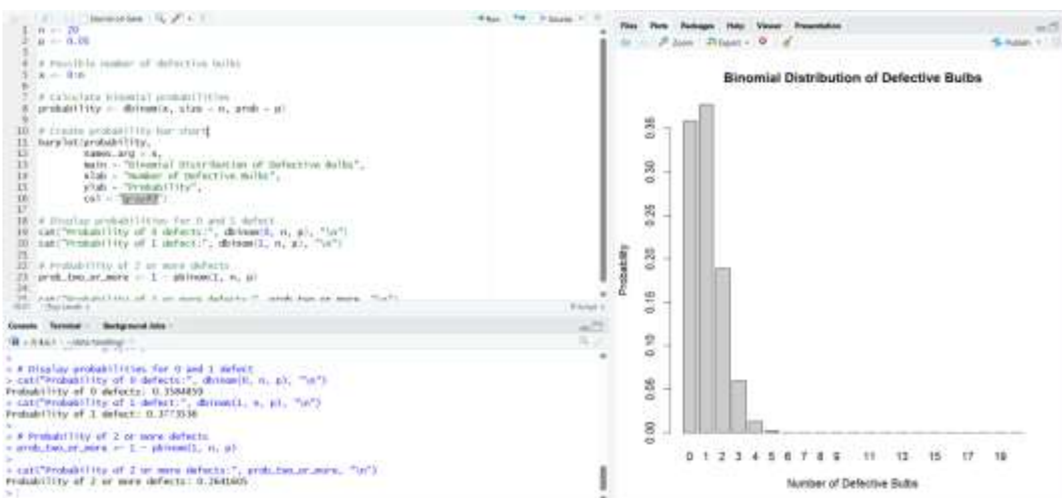
A bar chart is appropriate because we are showing the probability of discrete outcomes:

0 defects, 1 defect, 2 defects, etc.

Procedure:

1. Open RStudio.
2. Set the number of bulbs to 20.
3. Set the defect probability to 0.05.
4. Create possible defect counts from 0 to 20.
5. Use dbinom() to calculate the probability for each number of defects.
6. Create a bar chart.
7. Calculate the probability of 0 defects.
8. Calculate the probability of 1 defect.
9. Calculate the probability of 2 or more defects.
10. Run the program and observe the graph.

Output:



Interpretation:

The most likely outcome is **1 defective bulb**.

There is approximately a **35.85% chance of having no defective bulbs**.

There is approximately a **37.74% chance of having exactly one defective bulb**.

There is approximately a **26.41% chance of having two or more defective bulbs**.

Justification:

A binomial distribution is appropriate because each bulb has two possible outcomes:

- Defective
- Non-defective

The number of trials is fixed at 20, and the probability of a defect remains 5%.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks	
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions		
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation		
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis		
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers		
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps		

Faculty In-charge	Course Coordinator	Head of Department

Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________